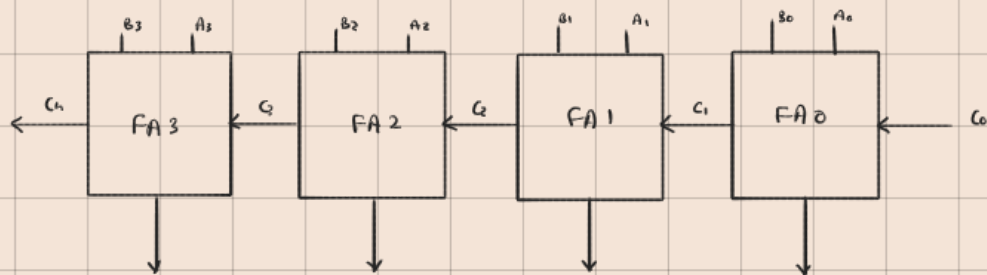


4 Bit Adder (Ripple Carry Adder)



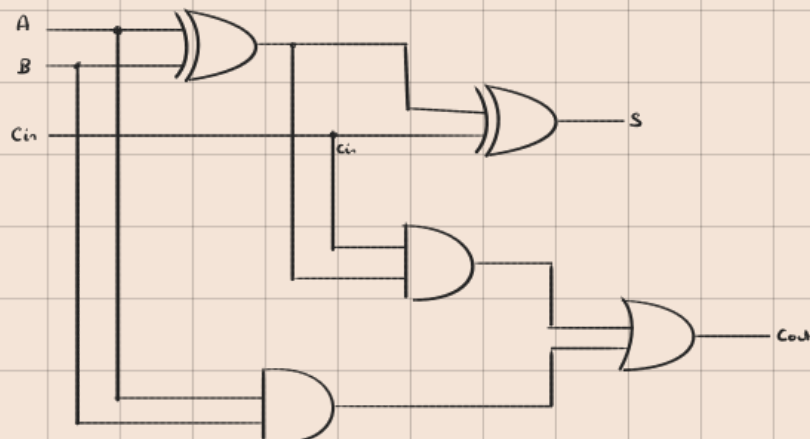
for an n bit adder

$\therefore t_c =$ time it takes to generate a carry

Delay for sum = $(n-1) t_c + t_s$

$\therefore t_s =$ time it takes to generate a sum

Delay for carry = $n t_c$



Carry Generation $\rightarrow C_i$

- Depends on the lower AND gate.
- Regardless of the upper cause of the OR gate.
- Carry is generated when both A and B is high cause of the AND gate.

$$C_i = A \cdot B$$

Carry Propagation $\rightarrow P$ (whether carry is passed to next gate or not)

- Upper AND gate.
- Both C_{in} and $A \oplus B$ must be 1
- Carry will propagate when A is 0 and B is 1 or vice versa.

$$P = A \oplus B$$

Carry Generation

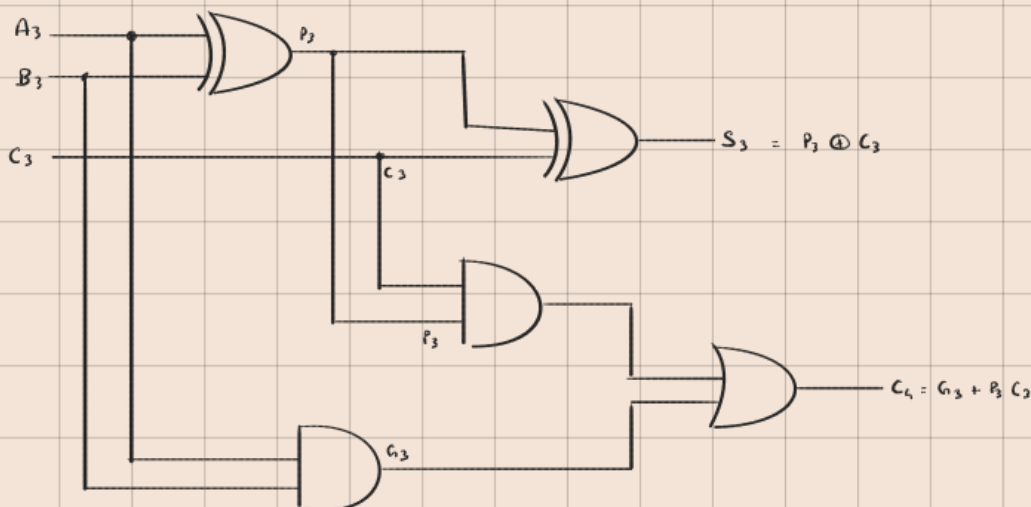
- $G_i = A_i \cdot B_i$ (AND gate)
- Both bits A and B must be 1
- Independent of Input Carry (C_{in})

- A new Carry is Created
- It passes a new generated carry regardless of what happened before it.

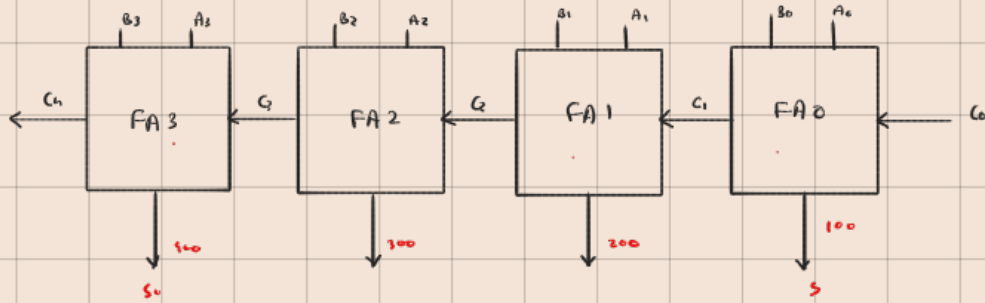
Carry Propagation

- $P_i = A_i \oplus B_i$ (XOR gate)
- Either bit A or B is 1
- Dependent on Input Carry (C_{in})
- Previous Carry is passed to next stage.

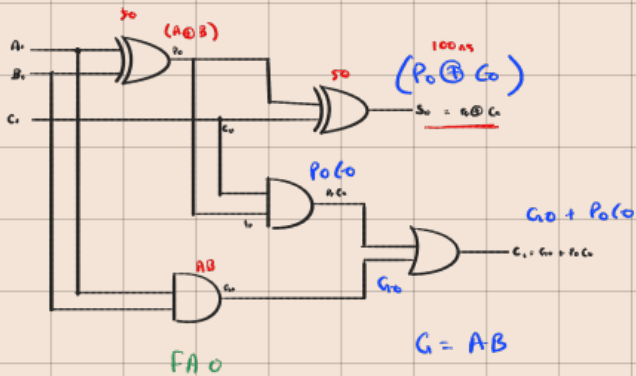
- It will only pass on a carry if it received a carry in from the previous stage.



4 bit Adder Using CLA

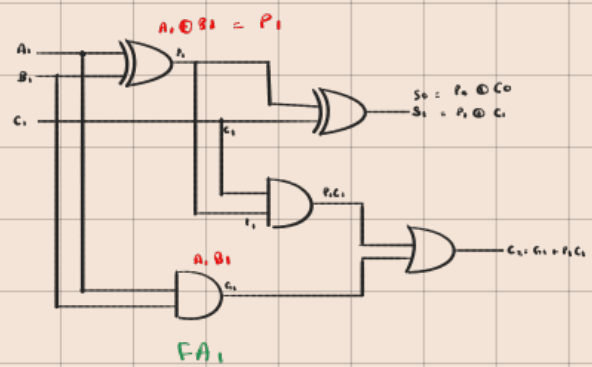


$$P_0 = A \oplus B$$



$$S_0 = P_0 \oplus C_0$$

$$C_1 = G_0 + P_0 C_0$$

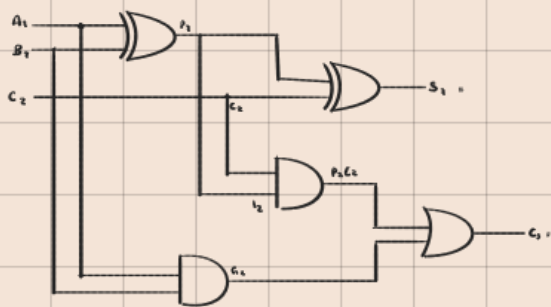


$$S_1 = P_1 \oplus C_1$$

$$C_2 = G_1 + P_1 C_1$$

$$C_2 = G_1 + P_1 (G_0 + P_0 C_0)$$

$$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$



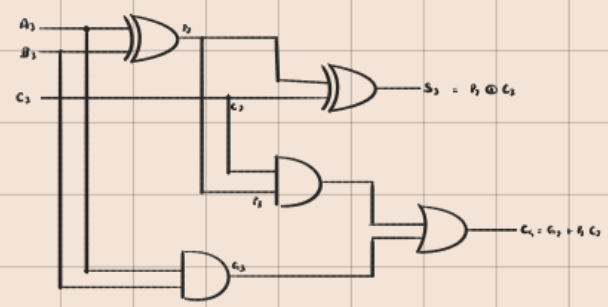
FA2

$$S_2 = P_2 \oplus C_2$$

$$C_3 = G_2 + P_2 C_2$$

$$= G_2 + P_2 (G_1 + P_1 G_0 + P_1 P_0 C_0)$$

$$C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$



FA3

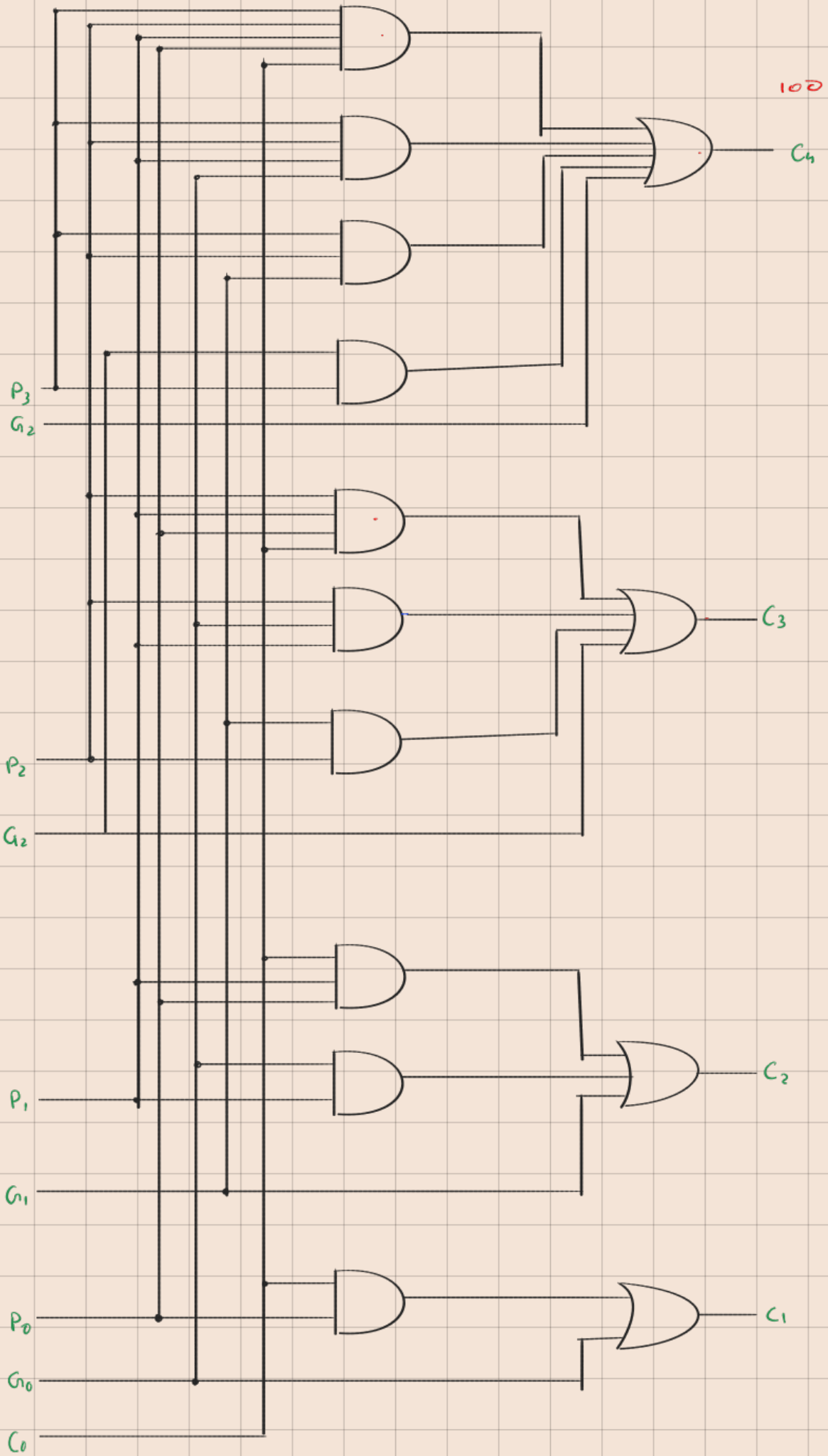
$$S_3 = P_3 \oplus C_3$$

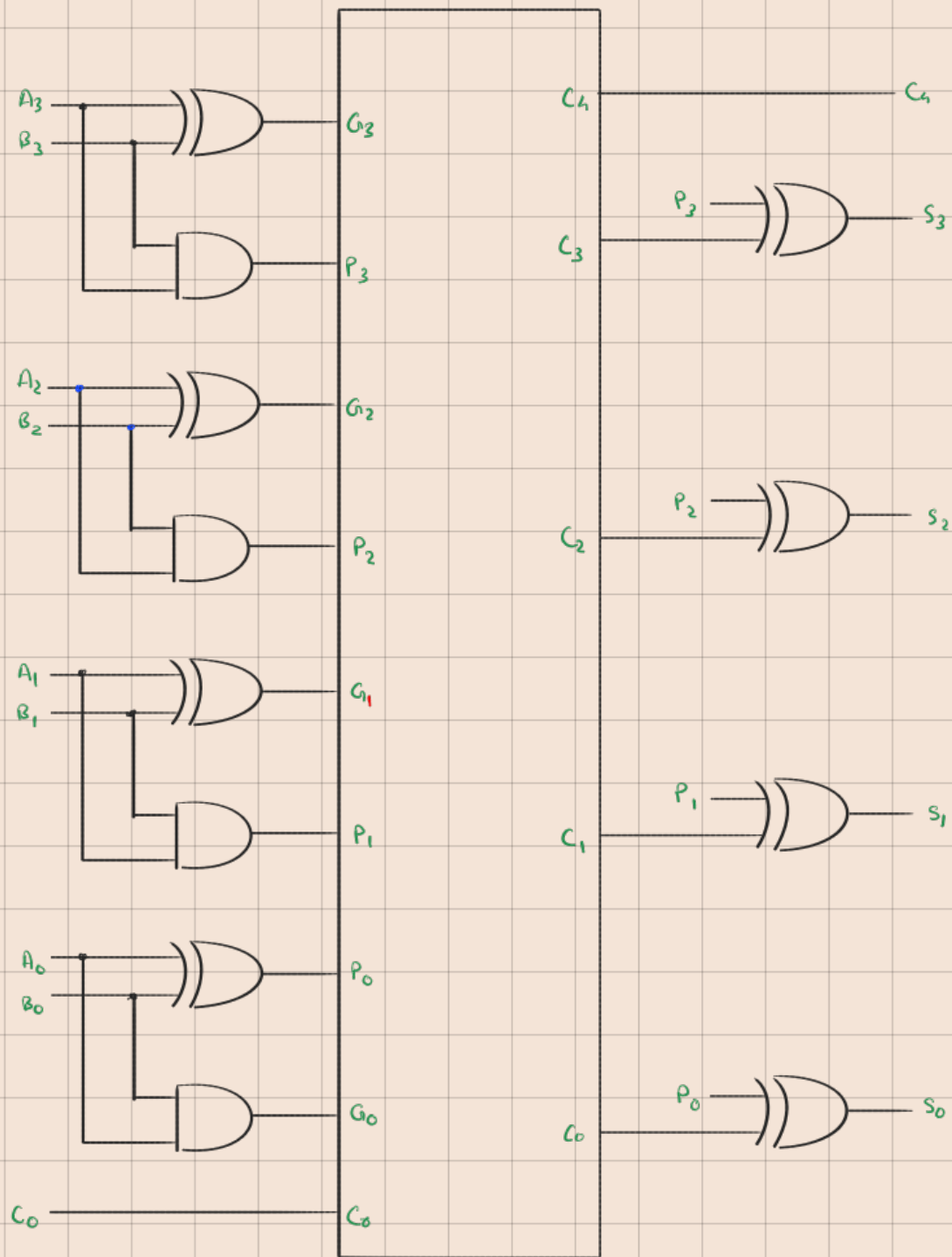
$$C_4 = G_3 + P_3 C_3$$

$$C_4 = G_3 + P_3 (G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0)$$

$$C_4 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$$

Full Combinational Circuit for the CLA

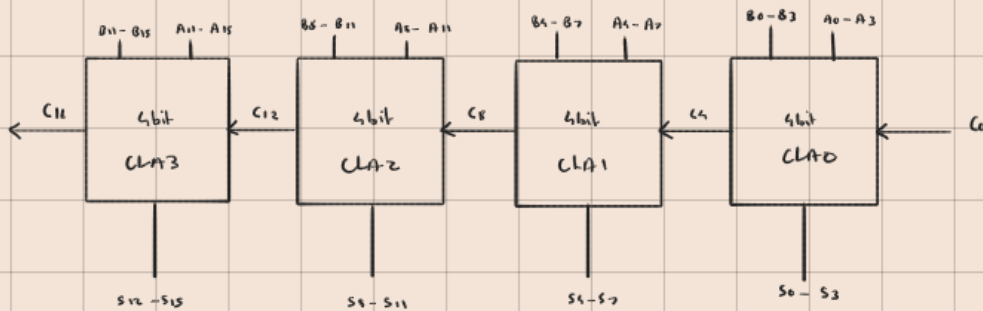




- total 4 gate delay
- instead of 9 gates in the ripple carry adder

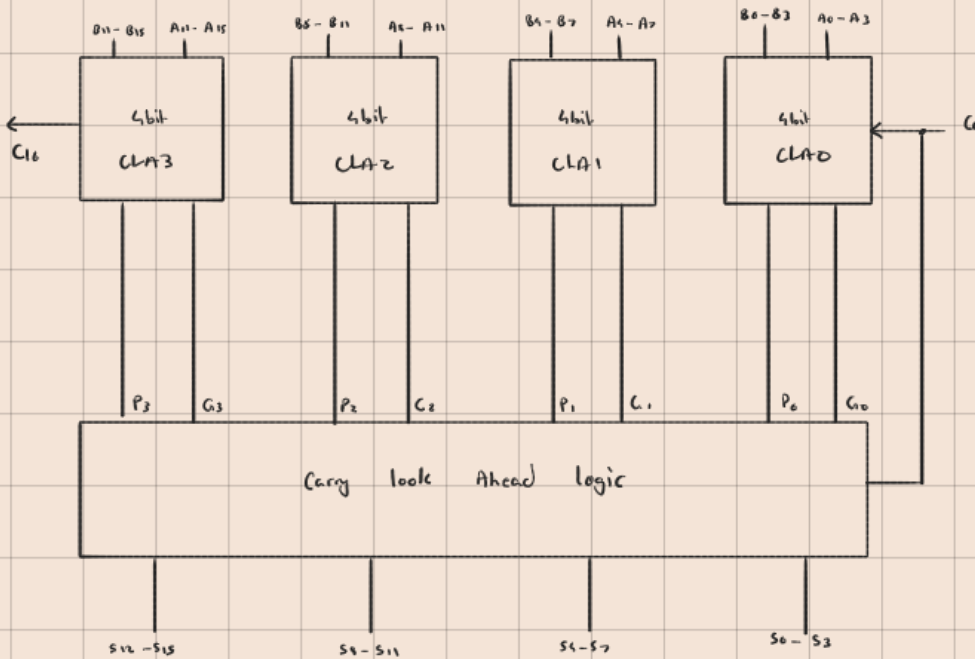
16 Bit Adder Using 4, 4 bits CLA

Using Ripple Carry



- 4, 4 bit CLA are added in a ripple Carry
- The last Carry from CLA ripples on to the next 4 bit CLA
- Finally carry delay is of 13 gates

Using Ripple Carry



Date _____

for C_2

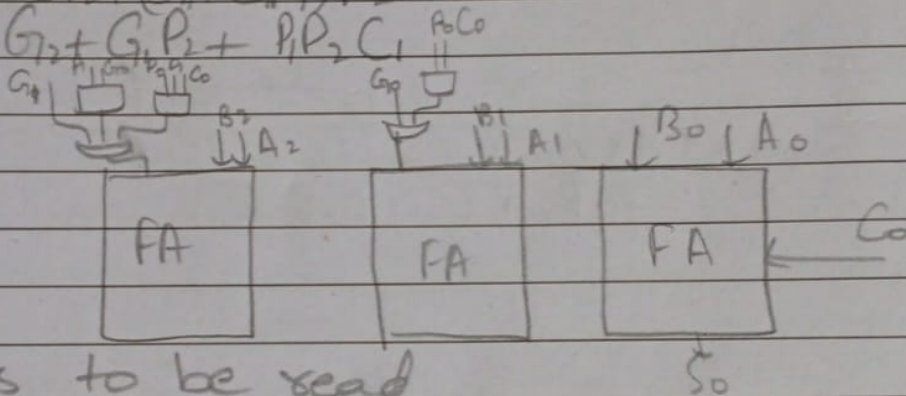
$$C_2 = G_1 + (G_0 + G_0 P_0) P_1$$

$$\Rightarrow C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$C_3 = C_2 + C_2 P_2$$

$$= C_2 + (G_1 + G_0 P_0) P_2$$

$$\Rightarrow C_3 = G_2 + G_1 P_1 + P_1 P_0 C_1$$



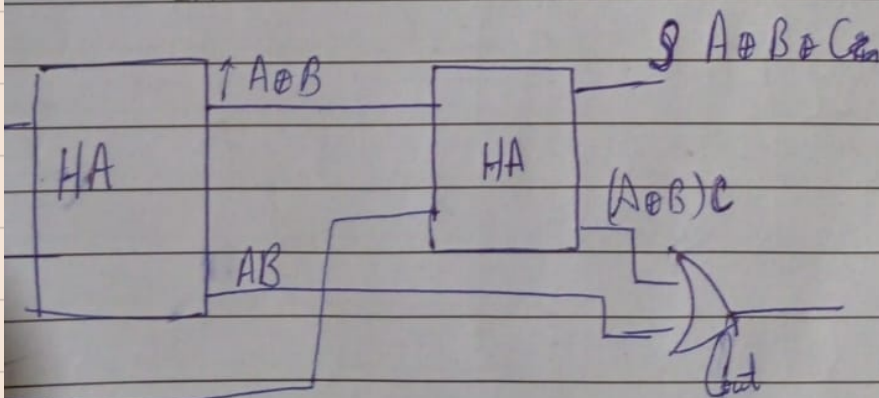
ICs to be used

7482

4008 / 7483

4582 / 74182

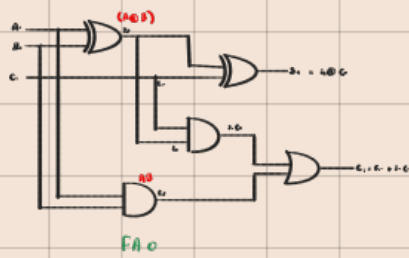
↳ CLA



$$(A \oplus B)C + AB$$

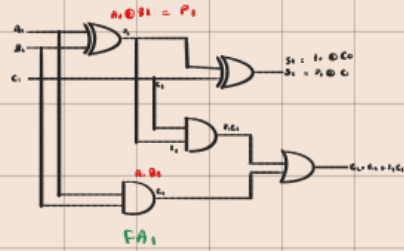
GENIUS

$$ABC + \bar{A}\bar{B}C + AB$$



$$S_0 = A_0 \oplus B_0$$

$$C_1 = A_0 B_0 + C_0 A_0$$

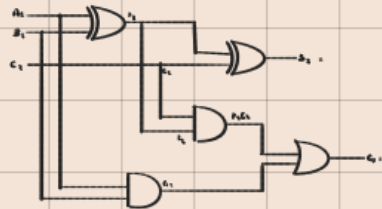


$$S_1 = A_1 \oplus B_1 \oplus C_0$$

$$C_2 = A_1 B_1 + C_1 A_1 + C_1 B_1$$

$$C_3 = C_2 + C_1 (A_1 B_1 + C_0 A_0 + C_0 B_0)$$

$$C_4 = C_3 + C_2 (A_1 B_1 + C_1 A_1 + C_1 B_1)$$

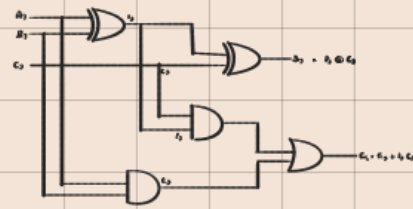


$$S_2 = A_2 \oplus B_2 \oplus C_1$$

$$C_3 = A_2 B_2 + C_2 A_2 + C_2 B_2$$

$$C_4 = C_3 + C_2 (A_2 B_2 + C_1 A_1 + C_1 B_1)$$

$$C_5 = C_4 + C_3 (A_2 B_2 + C_1 A_1 + C_1 B_1)$$

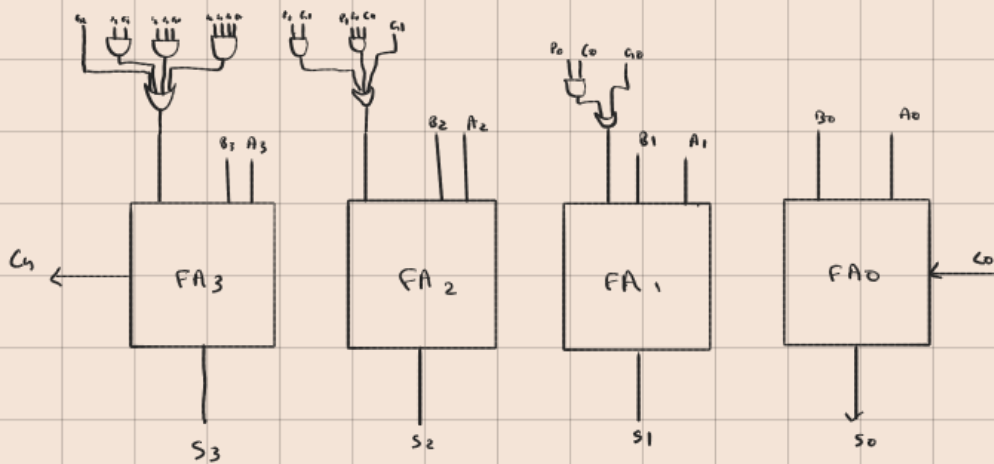


$$S_3 = A_3 \oplus B_3 \oplus C_2$$

$$C_4 = A_3 B_3 + C_3 A_3 + C_3 B_3$$

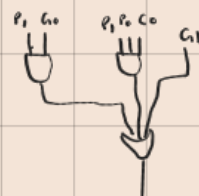
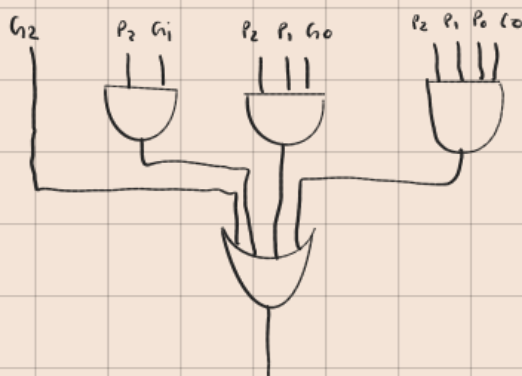
$$C_5 = C_4 + C_3 (A_3 B_3 + C_2 A_2 + C_2 B_2 + C_1 A_1 + C_1 B_1)$$

$$C_6 = C_5 + C_4 (A_3 B_3 + C_2 A_2 + C_2 B_2 + C_1 A_1 + C_1 B_1)$$



$$C_3 = C_2 + C_1 A_2 + C_1 B_2 + C_1 C_0$$

$$C_2 = C_1 + C_0 A_1 + C_0 B_1$$



$$G = A \cdot B$$

$$P = A \oplus B$$

$$C_1 = G_0 + P_0 C_0$$

$$C_2 = G_1 + P_1 (G_0 + P_0 C_0)$$

$$C_3 = G_2 + P_1 \underline{C_2}$$

